

MIT - Assignment - 2 (Lab)

27/08

Q1.

$$x + y + z = 6$$

$$2x + 3y + z = 10$$

$$3x + 4y + 2z = 15$$

The simplest way to check existence of solution would be to perform a gaussian elimination.

Augmented matrix, $(A|b) =$

$$\text{pivot-2} \rightarrow \begin{bmatrix} \textcircled{1} & 1 & 1 & | & 6 \\ 2 & 3 & 1 & | & 10 \\ 3 & 4 & 2 & | & 15 \end{bmatrix}$$

$$\begin{aligned} &\downarrow r_2' = r_2 - 2r_1 \\ &\downarrow r_3' = r_3 - 3r_1 \end{aligned}$$

$$\begin{aligned} &\text{pivot 2} \rightarrow \begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 0 & \textcircled{1} & -1 & | & -2 \\ 0 & 1 & -1 & | & -3 \end{bmatrix} \xrightarrow{r_3' = r_3 - r_2} \begin{bmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & -1 & | & -2 \\ 0 & 0 & 0 & | & -1 \end{bmatrix} \end{aligned}$$

Since row 3 basically says,

$$0 = -1,$$

→ No solution exists for the system of linear equations. (solution)

Q².

$$A = \begin{bmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

we know $A \cdot v = \lambda v$ [$\lambda \equiv$ eigenvalues ;
 $v \equiv$ eigenvector]

$$\Rightarrow (A - \lambda I) \cdot v = 0$$

$$\Rightarrow \det(A - \lambda I) = 0$$

$$\therefore \det \begin{pmatrix} 2-\lambda & -3 & 0 \\ 2 & -5-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{pmatrix} = 0$$

$$\Rightarrow (3-\lambda) [-(5+\lambda)(2-\lambda) + 6] = 0$$

$$\Rightarrow \lambda = 3 \quad \text{or} \quad \lambda^2 + 3\lambda - 4 = 0$$

$$\Rightarrow (\lambda + 4)(\lambda - 1) = 0$$

$$\Rightarrow \lambda = -4, \lambda = 1$$

$\therefore$ eigenvalues are $\lambda = 3, -4, 1$

(i) For $\lambda = 3$,

$$\begin{aligned} \cancel{2x} - x - 3y &= 0 \\ 2x - 8y &= 0 \\ 0 &= 0 \end{aligned} \quad \left[\text{Let } v = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \right]$$

From ①, $x = -3y$

$$\textcircled{2} \Rightarrow -6y - 8y = 0 \Rightarrow y = 0$$

$$\therefore x = 0$$

$$\text{Let } z = 1$$

$\therefore$ For eigenvalue $\lambda = 3$, eigenvector = $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

(ii) For $\lambda = -4$, let $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\therefore 6x - 3y = 0 \quad - (1)$$

$$2x - y = 0 \quad - (2)$$

$$7z = 0 \quad - (3)$$

$$\therefore z = 0 \quad \text{and} \quad y = 2x$$

$$\text{let } x = 1 \quad \text{and} \quad y = 2$$

$$\therefore \text{For } \lambda = -4, \text{ eigenvector} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

(iii) For $\lambda = 1$,

$$x - 3y = 0 \quad \therefore x = 3y$$

$$2x - 6y = 0 \quad z = 0$$

$$2z = 0 \quad \text{Let } x = 3, \quad y = 1$$

$$\therefore \text{For } \lambda = 1, \text{ eigenvector} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$$

Q3,

Input n

Create matrix A of size $n \times n$

Create vector b of size $n \times 1$

For $i = 1$ to n :

For $j = 1$ to n :

$A[i, j] = \text{Input}(\text{float})$

For $i = 1$ to n :

$b[i, 0] = \text{Input}(\text{float})$

Create augmented matrix $[A | b]$ (np.concat)

Find rank- A of matrix A

Find rank- Ab of matrix $[A | b]$

define rank(A):

set rank $= 0$

for each column:

Find a non-zero pivot downward
if a pivot row exists

swap the pivot row into current row

use pivot row to eliminate all entries
below pivot row.

rank ± 1

If $\text{rank } A = \text{rank } AB = n$,
print (unique solution exists)

For each column

Find a non-zero column and swap rows if necessary

Divide the pivot row by the pivot

For every other row below:

subtract a multiple of pivot row so that
the other entries in the column become 0

print (last column of AB (solution))

Else if $\text{rank } A = \text{rank } AB < n$,

print (infinitely many solutions exist)

Else if $\text{rank } A < \text{rank } AB$,

print (no solution exists)